Week 9

1. file_copy.c

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

int main(int argc, char *argv[])
{
  int source, destination;
  char buffer[1024];
  ssize_t bytes_read, bytes_written;
  long total_bytes = 0;

  if (argc != 3)
  {
    printf("Usage: %s <source> <destination>\n", argv[0]);
    return 1;
  }

  source = open(argv[1], O_RDONLY);

  if (source < 0)
  {
    perror("Error opening source file");
    return 1;
  }

  destination = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);

  if (destination < 0)
  {
    perror("Error opening destination file");
    close(source);
    return 1;
  }

  while ((bytes_read = read(source, buffer, sizeof(buffer))) > 0)
  {
    bytes_written = write(destination, buffer, bytes_read);
```

```
        if (bytes_written != bytes_read)
        {
            perror("Error writing file");
            close(source);
            close(destination);
            return 1;
        }

        total_bytes += bytes_written;
    }

    if (bytes_read < 0)
    {
        perror("Error reading file");
    }
    else
    {
        printf("File copied successfully.\n");
        printf("Bytes copied: %ld\n", total_bytes);
    }

    close(source);
    close(destination);

    return 0;
}
```

```
yaswanth6970@Yash6970:~/skill/week07$ mkdir -p ~/OSSP/Experiment9
cd ~/OSSP/Experiment9
yaswanth6970@Yash6970:~/OSSP/Experiment9$ nano file_copy.c
yaswanth6970@Yash6970:~/OSSP/Experiment9$ gcc -Wall -Wextra file_copy.c -o file_cop
y
yaswanth6970@Yash6970:~/OSSP/Experiment9$ printf 'Hello OSSP\nLinux System Calls us
ing WSL\n' > input.txt
yaswanth6970@Yash6970:~/OSSP/Experiment9$ ./file_copy input.txt output.txt
File copied successfully.
Bytes copied: 40
yaswanth6970@Yash6970:~/OSSP/Experiment9$ cat output.txt
cmp input.txt output.txt
echo $?
Hello OSSP
Linux System Calls using WSL
0
yaswanth6970@Yash6970:~/OSSP/Experiment9$
```

2. lib_copy.c

```
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *argv[])
{
  FILE *source;
  FILE *destination;
  char buffer[1024];
  size_t bytes_read;

  if (argc != 3)
  {
    printf("Usage: %s <source> <destination>\n", argv[0]);
    return 1;
  }

  source = fopen(argv[1], "rb");

  if (source == NULL)
  {
    perror("Error opening source file");
    return 1;
  }

  destination = fopen(argv[2], "wb");
```

```
    if (destination == NULL)
    {
        perror("Error opening destination file");
        fclose(source);
        return 1;
    }

    while ((bytes_read = fread(buffer, 1, sizeof(buffer), source)) > 0)
    {
        if (fwrite(buffer, 1, bytes_read, destination) != bytes_read)
        {
            perror("Error writing file");
            fclose(source);
            fclose(destination);
            return 1;
        }
    }

    printf("File copied successfully using standard library functions.\n");

    fclose(source);
    fclose(destination);

    return 0;
```

```
yaswanth6970@Yash6970:~/OSSP/Experiment9$ cat output.txt
cmp input.txt output.txt
echo $?
Hello OSSP
Linux System Calls using WSL
0
yaswanth6970@Yash6970:~/OSSP/Experiment9$ nano lib_copy.c
yaswanth6970@Yash6970:~/OSSP/Experiment9$ gcc -Wall -Wextra lib_copy.c -o lib_copy
./lib_copy input.txt library_output.txt
File copied successfully using standard library functions.
yaswanth6970@Yash6970:~/OSSP/Experiment9$ cat library_output.txt
cmp input.txt library_output.txt
echo $?
Hello OSSP
Linux System Calls using WSL
0
yaswanth6970@Yash6970:~/OSSP/Experiment9$
```

3. redirect_io.c

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
```

```
int main()
{
   int fd;

   fd = open("redirected_output.txt",
         O_WRONLY | O_CREAT | O_TRUNC,
         0644);
```

```
    if (fd < 0)
    {
        perror("Error opening output file");
        return 1;
    }

    if (dup2(fd, STDOUT_FILENO) < 0)
    {
        perror("dup2 failed");
        close(fd);
        return 1;
    }

    close(fd);

    printf("Hello OSSP\n");
    printf("Linux System Calls using WSL\n");

    return 0;
```

```
yaswanth6970@Yash6970:~/OSSP/Experiment9$ nano lib_copy.c
yaswanth6970@Yash6970:~/OSSP/Experiment9$ gcc -Wall -Wextra lib_copy.c -o lib_copy
./lib_copy input.txt library_output.txt
File copied successfully using standard library functions.
yaswanth6970@Yash6970:~/OSSP/Experiment9$ cat library_output.txt
cmp input.txt library_output.txt
echo $?
Hello OSSP
Linux System Calls using WSL
0
yaswanth6970@Yash6970:~/OSSP/Experiment9$ nano redirect_io.c
yaswanth6970@Yash6970:~/OSSP/Experiment9$ gcc -Wall -Wextra redirect_io.c -o redire
ct_io
./redirect_io
yaswanth6970@Yash6970:~/OSSP/Experiment9$ cat redirected_output.txt
Hello OSSP
Linux System Calls using WSL
yaswanth6970@Yash6970:~/OSSP/Experiment9$
```

```
}
```